Welcome to 151B: Deep Learning

Instructor: Annapurna Vadaparty

TA: Ben Bao

Tutors: Alexander Radulescu, Trevor Duong

Project Presentation Sign Ups

- Make sure you sign up for the correct length slot
- (Some of you are signed up incorrectly)
- Sign up by Friday (**or I will choose a spot for you)!**

Tues Jul 28	Presenter
11:00-11:05 - Attendance/welcome	
11:05-11:20 - Presentation 1	
11:20-11:25 - Q & A	
11:25-11:40 - Presentation 2	
11:40-11:45 - Q & A	
11:45-12:00 - Presentation 3	
12:00-12:05 - Q & A	
12:05-12:10 - Lightning Talk 1	
12:10-12:12 - Q & A	
12:12-12:17 - Lightning Talk 2	
12:17-12:19 - Q & A	

Monday July 27: Career Panel from Industry and Academia

Benjamin Stingle · 3rd

Product Leader

San Francisco, California, United States · Contact info

500+ connections

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About

A product leader with 10+ years experience building products, teams, and ventures, I've led product management from concept through launch several times, driven 15x annual growth, and closed several venture rounds. Aside from PM, have worn the full-stack engineer hat, have extensive experience as a venture investor, and have started a few ventures of my own.   I've works at organizations from two to 10k employees.

My experience spans a variety of domains, including: managed marketplaces, mental health and wellness, leadership development, misinformation, consumer and mobile software, enterprise SaaS, life science infrastructure, and cleantech. I've even experienced viral growth while crowdfunding America's first cat café.

I believe that great products are born from a deep understanding of and empathy with the user, rapid iteration, and tight collaboration between all stakeholders. As a product manager, I've designed product development processes to ensure that opportunities can demonstrate success quickly and while still building toward the long-term vision.

Also, ask me about the benefits of antifragility.

Fluid Intelligence

Fluid and crystallized intelligence

22 languages

Article Talk Read Edit View history

From Wikipedia, the free encyclopedia

The concepts of **fluid intelligence** (*g_f*) and **crystallized intelligence** (*g_c*) were introduced in 1943 by the psychologist Raymond Cattell.[1][2][3] According to Cattell's psychometrically-based theory, general intelligence (*g*) is subdivided into g_f and g_c. Fluid intelligence is the ability to solve novel reasoning problems. It is correlated with a number of important skills such as comprehension, problem-solving, and learning.[4] Crystallized intelligence, on the other hand, involves the ability to deduce secondary relational abstractions by applying previously learned primary relational abstractions.[5]

AGI and Fluid Intelligence

FOUNDATION

ARC Prize Foundation

A North Star for AGI

The ARC Prize Foundation is a non-profit dedicated to accelerating the development of Artificial General Intelligence (AGI). We believe that true AGI requires more than just scaling up existing AI models. It demands a fundamental shift towards systems capable of genuine fluid intelligence, the ability to adapt to new situations and solve novel problems efficiently, much like humans do.

Our core mission is to identify, measure, and ultimately close the capability gap between human and artificial intelligence on tasks that are simple for people yet remain difficult for even the most advanced AI systems today.

We achieve this by creating and curating human-calibrated benchmarks that serve as a clear and objective measure of progress towards AGI.

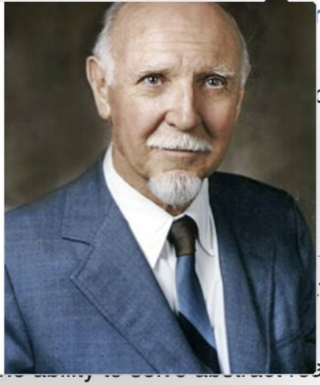
These benchmarks are not just tests, they are a driving force for AI research. They guide researchers towards the most important bottlenecks on the way to AGI, and incentivize them to explore approaches that go beyond pattern matching and memorization.

Think about a diverse range of fields and literature

History [edit]

Fluid and crystallized intelligence are constructs originally conceptualized by Raymond Cattell.[2] The concepts of fluid and crystallized intelligence were rn.[6][7][3] Most of the intelligence testing had mainly been focused on intelligence changed and developed when aging took place on ar ots remained, others diminished. Thus, there was a need to delineat

Raymond Bernard Cattell was a British-American psychologist, known for his psychometric research into intrapersonal psychological structure. His work also explored the basic dimensions of personality and temperament, the range of cognitive abilities, the dynamic



Fluid versus crysta

Fluid intelligence (g_f) invol t depend only minimally on prior learning (such as formal and informa can "flow into" a wide variety of cognitive activities.[9] Tasks easoning problems, for example figure classifications, figural analyses, number and letter series, matrices, and paired associates.[7]

Crystallized intelligence (g_c) is the application of learned procedures and knowledge, and depends heavily on experience and acculturation. Horn notes that crystallized ability is a "precipitate out of experience," resulting from the prior application of fluid ability that has been combined with the intelligence of culture.[9] Examples of tasks that measure crystallized intelligence are vocabulary, general information, abstract word analogies, and the mechanics of language.[7]

Monday July 27: Career Panel from Industry and Academia

Yuxuan (Effie) Li

I am a Research Scientist at Google DeepMind. I am broadly interested in machine and human cognition, and the mechanisms that support these abilities in deep learning models and the human brain. Some of my recent works include evaluating zero-shot visual reasoning in video models and understanding in-context task representations in language models.

My PhD research investigated a learning theory of task decomposition in transformers and humans. I also worked on multi-modal LLM evaluation and embodied AI as interns at Meta and AI2, and spent time building neural decoders of human episodic memory at UPenn.

Github | CV | Google Scholar | Linkedin

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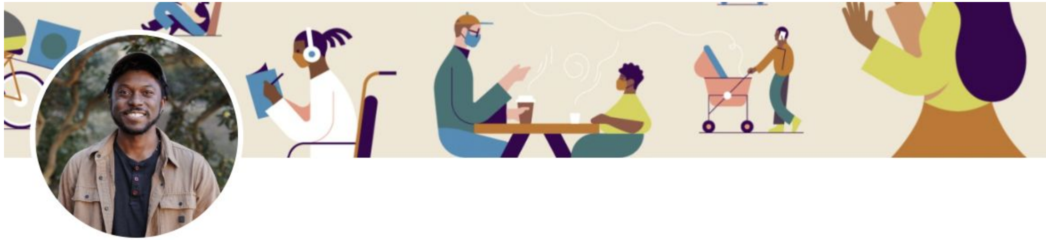

Ian Jackson He/Him · 2nd

PhD Student in Cognitive Science | GEM Fellow

San Diego, California, United States · **Contact info**

495 connections

Srinivas, **Quirine** and 8 other mutual connections

Connect Message ···

The Johns Hopkins University Applied Physics Laboratory

UC San Diego

About

I'm a PhD student in Cognitive Science at UCSD. Most of my job experience has been in software development for research organizations.

I'm interested in Brain-Computer Interfaces, Human-Computer Interaction, and Neural Engineering

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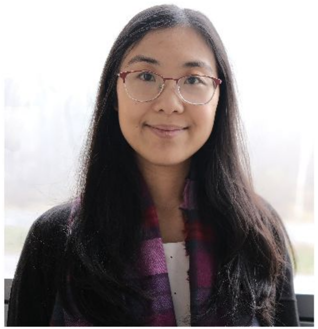
This is the homepage of Lisa Zhang.

Teaching . Publications . Undergrad Projects . Service . Contact . CV

Hi!



I am an Associate Professor, Teaching Stream at the Department of Mathematical and Computational Sciences (MCS), University of Toronto Mississauga. I hold non-budgetary cross-appointments at the Institute for the Study of University Pedagogy (ISUP) and at the Institute for Management and Innovation (IMI). My office is located at DH3072.

I have held many roles during my career: startup founder, data scientist, machine learning researcher, pure math student, and now a computer science educator and education researcher. I now mainly teach machine learning offerings at UTM. Previously, I have taught programming languages, data structures, introductory courses, and have led and coordinated the Writing Across the Curriculum (WAC) initiative for our computer science students.

Questions you'd be interested in asking them?

- What's it like to get a PhD?
- What are some things that surprised you after entering your field?
- Differences between working at a startup vs. big company?
  - (For different positions)
- At your current job, what's your favorite part of your day to day?
- How is doing research different in big tech companies vs. at universities?
- What are popular research topics in AI in industry?
- What are some of the main challenges you face in your job?
- Do you use AI to research AI?
- What are your thoughts on AI safety and recursive self improvement? What are your opinions on AI safety and effective altruism?
- What are current topics that you'd recommend for students to look into?
- What do you think of using AI to replace programmers? Do you think this is likely to happen?

Questions for me?

- Finding research positions:
    - Email grad students and postdocs, not just professors
    - Twitter can be useful–some people post there that they're looking for RAs
    - In general– "connections" can be people you already know

Guest Lecture from Prof. Cottrell

Garrison W. Cottrell

Courses **Pubs** **Fun** **Resources** **(Software, Data, my bio and vitae)** **Cognitive Modeling Greatest Hits**

Gary's Unbelievable Research Unit (very out of date!)
Computer Science & Engineering Department 0404
University of California, San Diego
La Jolla, CA 92093-0404 USA

E-mail: gary at-sign ucsd dot edu

My schedule (Be sure to use the week view to see the length of meetings)
My Google Scholar Page

Potential Post-docs, Grad students, summer interns, etc.:
I am **NOT** taking any new trainees at this time.
For advice on grad school, please read this!
For advice on women in tech, see this!

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I direct the Interdisciplinary Ph.D. Program (IDP) in Cognitive Science.
...and I direct(ed) the Temporal Dynamics of Learning Center.
Before emailing me about the IDP, please read this presentation on it that I wrote for incoming Neuroscience grads.
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